Table 1: Correction payloads, paths, and tags used in the analysis.

| Era | Correction | Payload / path | Key / tag |
|---|---|---|---|
| Run3_2022 | JEC/JER | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoAODv12/latest/jet_jerc.json.gz | JEC MC: Summer22_22Sep2023_V4_MC; JEC data: Summer22_22Sep2023_V4_DATA; JER: Summer22_22Sep2023_JRV2_MC |
| | JER smearing | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz | sources: JER, Total |
| | Pileup | /cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22CDSep23-Summer22-NanoAODv12/latest/puWeights.json.gz | Collisions2022_355100_357900_eraBCD_GoldenJson |
| | Muon SF | /cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22CDSep23-Summer22-NanoAODv12/latest/muon_Z.json.gz | NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight |
| | Muon ScaRe | corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoAODv12/muon_scalesmearing.json | standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoAODv12/muon_scalesmearing_VXBS.json |
| | Muon FSR | corrections/muon_fsr.py | algorithmic; no payload |
| | Jet veto map | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoAODv12/latest/jetvetomaps.json.gz | Summer22_23Sep2023_RunCD_V1 |
| | b-tag WP | /cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22CDSep23-Summer22-NanoAODv12/latest/btagging.json.gz | PNet_wp_values (L, M, T) |
| | Golden JSON | corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json | data luminosity mask |
| | Trigger SF | HLT_IsoMu24 | NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium |
| | LHE PDF uncertainty | LHEPdfWeight | replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity |
| | LHE QCD scales | LHEScaleWeight | muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip |
| Run3_2022EE | JEC/JER | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/jet_jerc.json.gz | JEC MC: Summer22EE_22Sep2023_V4_MC; JEC data: Summer22EE_22Sep2023_Run{}_V4_DATA, Summer22EE_22Sep2023_V4_DATA; JER: Summer22EE_22Sep2023_JRV2_MC |
| | JER smearing | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz | sources: JER, Total |
| | Pileup | /cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/puWeights.json.gz | Collisions2022_359022_362760_eraEFG_GoldenJson |
| | Muon SF | /cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/muon_Z.json.gz | NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight |
| | Muon ScaRe | corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoAODv12/muon_scalesmearing.json | standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoAODv12/muon_scalesmearing_VXBS.json |
| | Muon FSR | corrections/muon_fsr.py | algorithmic; no payload |
| | Jet veto map | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/jetvetomaps.json.gz | Summer22EE_23Sep2023_RunEFG_V1 |
| | b-tag WP | /cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/btagging.json.gz | PNet_wp_values (L, M, T) |
| | Golden JSON | corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json | data luminosity mask |
| | Trigger SF | HLT_IsoMu24 | NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium |
| | LHE PDF uncertainty | LHEPdfWeight | replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity |
| | LHE QCD scales | LHEScaleWeight | muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip |
| Run3_2023 | JEC/JER | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoAODv12/latest/jet_jerc.json.gz | JEC MC: Summer23Prompt23_V4_MC; JEC data: Summer23Prompt23_Run{}_V4_DATA, Summer23Prompt23_V4_DATA; JER: Summer23Prompt23_RunCv1234_JRV3_MC |
| | JER smearing | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz | sources: JER, Total |
| | Pileup | /cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23CSep23-Summer23-NanoAODv12/latest/puWeights.json.gz | Collisions2023_366403_369802_eraBC_GoldenJson |
| | Muon SF | /cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23CSep23-Summer23-NanoAODv12/latest/muon_Z.json.gz | NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight |
| | Muon ScaRe | corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoAODv12/muon_scalesmearing.json | standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoAODv12/muon_scalesmearing_VXBS.json |
| | Muon FSR | corrections/muon_fsr.py | algorithmic; no payload |
| | Jet veto map | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoAODv12/latest/jetvetomaps.json.gz | Summer23Prompt23_RunC_V1 |
| | b-tag WP | /cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23CSep23-Summer23-NanoAODv12/latest/btagging.json.gz | PNet_wp_values (L, M, T) |
| | Golden JSON | corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json | data luminosity mask |
| | Trigger SF | HLT_IsoMu24 | NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium |
| | LHE PDF uncertainty | LHEPdfWeight | replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity |

Continued on next page

Table 1 – continued

| Era | Correction | Payload / path | Key / tag |
|---|---|---|---|
| | LHE QCD scales | LHEScaleWeight | muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip |
| Run3_2023BPix | JEC/JER | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jet_jerc.json.gz | JEC MC: Summer23BPixPrompt23_V4_MC; JEC data: Summer23BPixPrompt23_Run{}_V4_DATA, Summer23BPixPrompt23_V4_DATA; JER: Summer23BPixPrompt23_RunD_JRV3_MC |
| | JER smearing | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz | sources: JER, Total |
| | Pileup | /cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/puWeights.json.gz | Collisions2023_369803_370790_eraD_GoldenJson |
| | Muon SF | /cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/muon_Z.json.gz | NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight |
| | Muon ScaRe | corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoAODv12/muon_scalesmearing.json | standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoAODv12/muon_scalesmearing_VXBS.json |
| | Muon FSR | corrections/muon_fsr.py | algorithmic; no payload |
| | Jet veto map | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jetvetomaps.json.gz | Summer23BPixPrompt23_RunD_V1 |
| | b-tag WP | /cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/btagging.json.gz | PNet_wp_values (L, M, T) |
| | Golden JSON | corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json | data luminosity mask |
| | Trigger SF | HLT_IsoMu24 | NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium |
| | LHE PDF uncertainty | LHEPdfWeight | replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity |
| | LHE QCD scales | LHEScaleWeight | muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip |
| Run3_2024 | JEC/JER | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz | JEC MC: Summer24Prompt24_V5_MC; JEC data: Summer24Prompt24_V5_DATA; JER: Summer24Prompt24_JRV2_MC |
| | JER smearing | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz | sources: JER, Total |
| | Pileup | /cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/puWeights_BCDEFGHI.json.gz | Collisions24_BCDEFGHI_goldenJSON |
| | Muon SF | /cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/muon_Z.json.gz | NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight |
| | Muon ScaRe | corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing.json | standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing_ |
| | Muon FSR | corrections/muon_fsr.py | algorithmic; no payload |
| | Jet veto map | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz | Summer24Prompt24_RunBCDEFGHI_V1 |
| | b-tag WP | /cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/btagging.json.gz | UParTAK4_wp_values (L, M, T) |
| | Golden JSON | corrections/data/golden_json/Cert_Collisions2024_378981_386951_Golden.json | data luminosity mask |
| | Trigger SF | HLT_IsoMu24 | NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium |
| | LHE PDF uncertainty | LHEPdfWeight | replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity |
| | LHE QCD scales | LHEScaleWeight | muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip |
| Run3_2025 | JEC/JER | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz | JEC MC: Summer24Prompt25_V3_MC; JEC data: Summer24Prompt25_V3_DATA; JER: Summer24Prompt25_JRV2_MC |
| | JER smearing | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz | sources: JER, Total |
| | Pileup | /cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz | Collisions25_goldenJSON |
| | Muon SF | /afs/cern.ch/work/v/vdamante/Hmm_newSkim/corrections/data/MUO/SF/Run3-25Prompt-Summer24-NanoAODv15/muon_Z.json.gz | NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight |
| | Muon ScaRe | corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing.json | standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json |
| | Muon FSR | corrections/muon_fsr.py | algorithmic; no payload |
| | Jet veto map | /cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz | Summer24Prompt25_RunCDEFG_V1 |
| | b-tag WP | /cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-25Prompt-Summer24-NanoAODv15/latest/btagging.json.gz | UParTAK4_wp_values (L, M, T) |
| | Golden JSON | corrections/data/golden_json/Cert_Collisions2025_391658_398903_Golden.json | data luminosity mask |
| | Trigger SF | HLT_IsoMu24 | NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium |

Continued on next page

Table 1 – continued

| Era | Correction | Payload / path | Key / tag |
|---|---|---|---|
| | LHE PDF uncertainty | `LHEPdfWeight` | `replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity` |
| | LHE QCD scales | `LHEScaleWeight` | `muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip` |
| Run3_2026 | JEC/JER | `/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz` | `JEC MC: Summer24Prompt26_V2_MC; JEC data: Summer24Prompt26_Run{}_V1_DATA, Summer24Prompt26_V1_DATA; JER: Summer24Prompt25_JRV2_MC` |
| | JER smearing | `/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz` | `sources: JER, Total` |
| | Pileup | `/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz` | `Collisions25_goldenJSON` |
| | Muon SF | `/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-26Prompt-Summer24-NanoAODv15/latest/muon_Z.json.gz` | |
| | Muon ScaRe | `corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing.json` | `standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json` |
| | Muon FSR | `corrections/muon_fsr.py` | `algorithmic; no payload` |
| | Jet veto map | `/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz` | `Summer24Prompt26_RunBCD_V1` |
| | b-tag WP | `/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-26Prompt-Summer24-NanoAODv15/latest/btagging.json.gz` | `UParTAK4_wp_values (L, M, T)` |
| | Golden JSON | `corrections/data/golden_json/Cert_Collisions2026_401624_403937_golden.json` | `data luminosity mask` |
| | Trigger SF | `HLT_IsoMu24` | `NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium` |
| | LHE PDF uncertainty | `LHEPdfWeight` | `replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity` |
| | LHE QCD scales | `LHEScaleWeight` | `muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip` |

**LHE PDF uncertainty.** The nominal member is `LHEPdfWeight[0]` and replicas 1–100 are used. For each event, the relative replica RMS is $\sigma_{\mathrm{PDF}} = \sqrt{\sum_i (w_i/w_0 - 1)^2/(N_{\mathrm{valid}} - 1)}$. The multiplicative variations are $w_{\mathrm{PDF}}^{\mathrm{Up}} = 1 + \sigma_{\mathrm{PDF}}$ and $w_{\mathrm{PDF}}^{\mathrm{Down}} = \max(0, 1 - \sigma_{\mathrm{PDF}})$. The configured alphaS members are not included.

**LHE QCD-scale uncertainty.** $\mu_R$ is the QCD renormalization scale, which controls the scale at which $\alpha_s$ is evaluated. $\mu_F$ is the factorization scale, which separates long-distance PDF evolution from the short-distance hard process. Values 0.5, 1, and 2 are multiplicative factors relative to the nominal generator scales. The configured final nuisances are: `QCD_ren_scale`: Down=$(\mu_R, \mu_F) = (0.5, 1)$ (`LHEScaleWeight[1]`), Up=$(2, 1)$ (index 7); and `QCD_fac_scale`: Down=$(1, 0.5)$ (index 3), Up=$(1, 2)$ (index 5). The corner points $(0.5, 0.5)$ (index 0) and $(2, 2)$ (index 8) are listed and normalized in the skim metadata but are not used by the two final Up/Down pairs. Each scale template is normalized with its own dataset-wide denominator from the skim reports before histogram filling.